

Disproof of conjecture 00000003955

earthking11

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Verdict: FALSE

We disprove conjecture 00000003955 as stated. The conjecture is built on the assertion that rank-width r and treewidth t always satisfy the “classical inequality”

$$t \leq 3r - 1. \quad (1)$$

That inequality is already false, and it fails at the smallest non-trivial complete graph: K_4 has rank-width 1 and treewidth 3, so

$$t = 3 > 2 = 3 \cdot 1 - 1 = 3r - 1.$$

Because the claimed inequality is false, the follow-up assertions of the conjecture (that the constant 3 is optimal, and that a family of rank-width r and treewidth exactly $3r - 1$ exists) cannot be established on the stated basis. In fact the situation is far worse than a wrong constant: we show that

$$\text{rw}(K_n) = 1, \quad \text{tw}(K_n) = n - 1 \quad (n \geq 2),$$

so the treewidth of a graph is *not bounded by any function of its rank-width*. The constant 3 is therefore neither optimal nor even meaningful: no inequality of the form $t \leq f(r)$ can hold. The smallest counterexample is K_4 ; the inequality $t \leq 3r - 1$ happens to hold for K_2 ($1 \leq 2$) and is saturated at K_3 ($2 = 3 \cdot 1 - 1$), but fails for every K_n with $n \geq 4$.

1 The conjecture

Quoted verbatim from `conjectures/00000003955.md`:

English. Definition: The rank-width is a width parameter satisfying, with treewidth t , the classical inequality $t \leq 3r - 1$. Conjecture: The constant 3 in the inequality $t \leq 3r - 1$ is optimal, with an explicit family of graphs of rank-width r and treewidth exactly $3r - 1$; moreover the kernelization kernel size for distance-hereditary graphs is constant. (optimal rank-width-treewidth constant)

中文。定义: rank-width 指宽度参数, 与树宽满足 $t \leq 3r - 1$ 的经典不等式。猜想: 不等式 $t \leq 3r - 1$ 中的常数 3 最优, 存在秩宽 r 树宽恰 $3r - 1$ 的显式图族; 且距离遗传图的核化核尺寸为常数。(秩宽-树宽常数最优)

The conjecture contains a definitional premise, namely (1), and two claims resting on it. Refuting the premise refutes the conjecture: a conjecture whose stated definition is false cannot be correct. We therefore concentrate on (1).

2 Definitions

Definition 1 (Cut-rank over $\text{GF}(2)$). Let $G = (V, E)$ be a finite simple graph and let (X, Y) be a bipartition of V (so $X \cap Y = \emptyset$ and $X \cup Y = V$). The *cut matrix* of (X, Y) is the $|X| \times |Y|$ matrix $A = A(G, X, Y)$ over $\text{GF}(2)$ with

$$A_{xy} = \begin{cases} 1 & \text{if } xy \in E, \\ 0 & \text{otherwise,} \end{cases} \quad x \in X, y \in Y.$$

The *cut-rank* of (X, Y) is $\text{cutrk}_G(X, Y) = \text{rank}_{\text{GF}(2)} A(G, X, Y)$.

Definition 2 (Rank-width [1, 2]). A *rank-decomposition* of G is a pair (T, τ) where T is a binary tree with $|V|$ leaves and $\tau: V \rightarrow L(T)$ is a bijection from the vertices to the leaves. For an edge e of T , deleting e splits the leaves into two classes X_e, Y_e , and hence splits V into a bipartition (X_e, Y_e) . The *width* of (T, τ) is

$$\max_{e \in E(T)} \text{cutrk}_G(X_e, Y_e),$$

and the *rank-width* of G is

$$\text{rw}(G) = \min_{(T, \tau)} \max_{e \in E(T)} \text{cutrk}_G(X_e, Y_e),$$

the minimum being over all rank-decompositions of G .

Definition 3 (Tree decomposition and treewidth). A *tree decomposition* of $G = (V, E)$ is a pair $(T, \{B_t\}_{t \in V(T)})$, where T is a tree and each *bag* B_t is a subset of V , such that:

- (i) $\bigcup_t B_t = V$;
- (ii) for every edge $uv \in E$ there is a node t with $u, v \in B_t$;
- (iii) for every $v \in V$, the set $\{t : v \in B_t\}$ induces a connected subtree of T .

The *width* of the decomposition is $\max_t |B_t| - 1$, and the *treewidth* $\text{tw}(G)$ is the minimum width over all tree decompositions of G .

Remark 4. The two parameters are of very different natures. Rank-width is a *min-max* parameter over all bipartitions arising in rank-decompositions, while treewidth is a min-max parameter over tree decompositions. That the two can be arbitrarily far apart is exactly what the complete graphs exhibit.

3 Rank-width of K_4 is 1

Let K_4 have vertex set $V = \{0, 1, 2, 3\}$, with $ij \in E$ for all distinct i, j .

Lemma 5 (Cuts of a complete graph). *Let $n \geq 2$ and let (X, Y) be a bipartition of $V(K_n)$ with $X, Y \neq \emptyset$. Then every entry of the cut matrix $A(K_n, X, Y)$ equals 1, and*

$$\text{rank}_{\text{GF}(2)} A(K_n, X, Y) = 1.$$

Proof. For $x \in X$ and $y \in Y$ we have $x \neq y$ because X and Y are disjoint, hence $xy \in E(K_n)$; therefore $A_{xy} = 1$ for every pair (x, y) . Write $\mathbf{1}_X \in \text{GF}(2)^{|X|}$ and $\mathbf{1}_Y \in \text{GF}(2)^{|Y|}$ for the all-ones column vectors. Then $A = \mathbf{1}_X \mathbf{1}_Y^\top$ is an outer product, so $\text{rank } A \leq 1$. Since $A \neq 0$ (both sides are nonempty, so A has at least one entry, and that entry is 1), $\text{rank } A \geq 1$. Hence $\text{rank } A = 1$. $\square$

Proposition 6. $\text{rw}(K_4) = 1$.

Proof. Upper bound $\text{rw}(K_4) \leq 1$. The seven bipartitions of a four-element set, up to swapping the two sides, are

$$\begin{aligned} \{0\}|\{1, 2, 3\}, \quad \{1\}|\{0, 2, 3\}, \quad \{2\}|\{0, 1, 3\}, \quad \{3\}|\{0, 1, 2\}, \\ \{0, 1\}|\{2, 3\}, \quad \{0, 2\}|\{1, 3\}, \quad \{0, 3\}|\{1, 2\}. \end{aligned}$$

By Lemma 5, every one of these has cut-rank 1. Now take the *caterpillar* rank-decomposition of K_4 that splits off one vertex at a time. Its internal edges induce precisely the cuts

$$\{0\} | \{1, 2, 3\}, \quad \{1\} | \{2, 3\}, \quad \{2\} | \{3\},$$

each of cut-rank 1 by Lemma 5. Hence this rank-decomposition has width 1, so $\text{rw}(K_4) \leq 1$.

Lower bound $\text{rw}(K_4) \geq 1$. Let (T, τ) be any rank-decomposition of K_4 . Its root edge splits V into two nonempty classes X, Y (a binary tree with $|V| \geq 2$ leaves has a nontrivial root split), and this bipartition is one of the seven above, so by Lemma 5 it has cut-rank 1. Therefore the width of (T, τ) is at least 1; since (T, τ) was arbitrary, $\text{rw}(K_4) \geq 1$.

Combining, $\text{rw}(K_4) = 1$. □

Remark 7. The same argument gives $\text{rw}(K_n) \leq 1$ for every $n \geq 2$ (split off one vertex at a time; every cut is a complete bipartite cut, of rank 1) and $\text{rw}(K_n) \geq 1$ because K_n has an edge and a rank-decomposition of width 0 would force every cut matrix to vanish, i.e. an edgeless graph. Hence $\text{rw}(K_n) = 1$ for all $n \geq 2$.

4 Treewidth of K_4 is 3

Lemma 8 (Minimum-degree bound). *Let G be a graph with at least one vertex and $\text{tw}(G) \leq w$. Then G has a vertex of degree at most w .*

Proof. Let $(T, \{B_t\})$ be a tree decomposition of G of width at most w . We first reduce the decomposition so that no leaf bag is contained in its neighbour's bag. If t is a leaf of T and $B_t \subseteq B_{t'}$ for its unique neighbour t' , delete t from T : conditions (i) and (ii) of Definition 3 remain true because every vertex of B_t is also in $B_{t'}$, and condition (iii) remains true because removing a leaf from a connected subtree leaves it connected; the width does not increase. Repeating this finitely many times (each step removes a node) yields a decomposition of width at most w in which no leaf bag is contained in its neighbour's bag (and in the base case where T has a single node, the argument below applies directly).

If T has a single node t , then $B_t = V$ by condition (i), so for any vertex v all its neighbours lie in B_t and $\deg(v) \leq |B_t| - 1 \leq w$.

Otherwise T has at least two nodes. Let t be a leaf and t' its neighbour. By the reduction, $B_t \not\subseteq B_{t'}$, so choose $v \in B_t \setminus B_{t'}$. We claim v lies in no other bag: by condition (iii) the bags containing v form a connected subtree T_v of T containing the leaf t ; a connected subtree containing a leaf either is the singleton $\{t\}$ or contains the neighbour t' . The second alternative is impossible because $v \notin B_{t'}$. Hence $T_v = \{t\}$. Now let u be any neighbour of v ; by condition (ii) some bag contains both u and v , and since the only bag containing v is B_t , we get $u \in B_t$. Thus every neighbour of v lies in B_t , so

$$\deg(v) \leq |B_t| - 1 \leq w. \quad \square$$

Corollary 9. *For every graph G with at least one vertex, $\text{tw}(G) \geq \delta(G)$, where $\delta(G)$ is the minimum degree of G .*

Proof. If $\text{tw}(G) \leq w$ then by Lemma 8 some vertex has degree $\leq w$; in particular $w \geq \delta(G)$. Taking $w = \text{tw}(G)$ gives $\text{tw}(G) \geq \delta(G)$. $\square$

Proposition 10. $\text{tw}(K_4) = 3$.

Proof. Upper bound $\text{tw}(K_4) \leq 3$. The tree T with a single node t and the single bag $B_t = \{0, 1, 2, 3\}$ is a tree decomposition: (i) holds since $B_t = V$; (ii) holds since both endpoints of every edge lie in B_t ; (iii) holds trivially. Its width is $|B_t| - 1 = 3$. Hence $\text{tw}(K_4) \leq 3$.

Lower bound $\text{tw}(K_4) \geq 3$. Every vertex of K_4 is adjacent to the other three, so $\delta(K_4) = 3$. By Corollary 9, $\text{tw}(K_4) \geq \delta(K_4) = 3$.

Combining, $\text{tw}(K_4) = 3$. $\square$

5 The inequality $t \leq 3r - 1$ fails

Theorem 11. *Conjecture 00000003955 is false. The graph K_4 has rank-width $r = \text{rw}(K_4) = 1$ and treewidth $t = \text{tw}(K_4) = 3$, and*

$$t = 3 > 2 = 3 \cdot 1 - 1 = 3r - 1.$$

Thus the stated “classical inequality” $t \leq 3r - 1$ is false, and with it the conjecture that rests on it.

Proof. Combine Proposition 6 ($r = 1$), Proposition 10 ($t = 3$), and $3 > 3 \cdot 1 - 1 = 2$. $\square$

Theorem 12 (No function bounds treewidth by rank-width). *For every $n \geq 2$,*

$$\text{rw}(K_n) = 1, \quad \text{tw}(K_n) = n - 1.$$

Consequently there is no function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $\text{tw}(G) \leq f(\text{rw}(G))$ for all graphs G .

Proof. The rank-width statement is the remark following Proposition 6. For the treewidth: the single bag $\{0, \dots, n - 1\}$ gives $\text{tw}(K_n) \leq n - 1$, while every vertex has degree $n - 1$, so Corollary 9 gives $\text{tw}(K_n) \geq n - 1$; hence $\text{tw}(K_n) = n - 1$. Now let f be any function and choose $n > f(1) + 1$. Then $\text{rw}(K_n) = 1$ and $\text{tw}(K_n) = n - 1 > f(1) = f(\text{rw}(K_n))$. So no such f exists. $\square$

Corollary 13 (Smallest counterexample and saturation). *The inequality $t \leq 3r - 1$ holds at K_2 ($1 \leq 2$) and is saturated at K_3 ($2 = 3 \cdot 1 - 1$), and fails for every K_n with $n \geq 4$. In particular K_4 is the smallest counterexample to the stated inequality.*

Proof. For K_n we have $r = 1$ and $t = n - 1$, so the inequality reads $n - 1 \leq 2$, i.e. $n \leq 3$. It holds exactly for $n \in \{2, 3\}$ (with equality at $n = 3$) and fails for all $n \geq 4$. $\square$

Remark 14. The conjecture’s final clause, “the kernelization kernel size for distance-hereditary graphs is constant”, is not needed for the refutation: the conjecture asserts the false inequality (1) as part of its definition, so it is false as stated irrespective of that clause. We make no claim about the kernelization clause here.

6 Verification and reproducibility

The finite claims are checked by a dependency-free Python 3 program `reproduce.py` (standard library only). For $n = 2, \dots, 7$ it:

- enumerates all $2^{n-1} - 1$ bipartitions of K_n , forms each cut matrix over $\text{GF}(2)$, computes its rank by Gaussian elimination, and reports the maximum (equal to 1; for K_4 all seven bipartitions have rank 1);
- verifies that the caterpillar rank-decomposition has all cuts of rank 1, witnessing $\text{rw}(K_n) \leq 1$, together with the lower bound $\text{rw}(K_n) \geq 1$;
- computes $\text{tw}(K_n)$ exactly as the minimum elimination width over all $n!$ elimination orders (the standard elimination-game characterisation of treewidth), verifies the minimum-degree bound $\text{tw} \geq \delta$, and exhibits the width- $(n - 1)$ decomposition;
- tabulates t against $3r - 1$, asserts that the inequality fails for every $n \geq 4$, and prints `PASS/FAIL` with a non-zero exit status on failure.

For K_4 it reports $r = 1$, $t = 3$, $3r - 1 = 2$, and $3 > 2$.

A Lean 4 formalisation accompanies this document in `lean4/`. It is written in core Lean only (`import Std`, no `Mathlib`) and contains no `sorry`. It formalises a computable $\text{GF}(2)$ rank of a matrix given by bit-mask rows, the cut matrix of a bipartition of K_4 , the fact that all seven bipartitions of K_4 have cut-rank 1, a lightweight treewidth model on the vertex type `Fin 4` (the elimination-order characterisation), the minimum-degree lemma, the explicit width-3 decomposition of K_4 , the computed value $\text{tw}(K_4) = 3$, and the conclusion $\text{tw}(K_4) > 3 \cdot \text{rw}(K_4) - 1$. See `lean4/README.md` for the precise scope of the formalisation.

References

- [1] S. Oum and P. Seymour. *Approximating clique-width and branch-width*. J. Combin. Theory Ser. B **96** (2006), 514–528.
- [2] S. Oum. *Rank-width and tree-width of claw-free graphs*. Discrete Math. **309** (2009), 2887–2889.